Module No. 02Fundamentals of Logical Thinking - II

1. The Eight Directions and Their Names

The eight primary directions are an extension of the four cardinal points. They are equally spaced at 45° angles from each other.

- Cardinal Directions: North (N), East (E), South (S), West (W).
- Ordinal (Intercardinal) Directions: Northeast (NE), Southeast (SE), Southwest (SW), Northwest (NW).

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          North (0°/360°)
              |
              |
West (270°) --+-- East (90°)
              |
              |
          South (180°)
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The ordinals sit exactly between them:

- NE is between N and E (45°)
- SE is between S and E (135°)
- SW is between S and W (225°)
- NW is between N and W (315°)

2. Angles Between Directions

Understanding the angular relationship is crucial for solving problems.

- Angle between adjacent directions (e.g., N and NE, NE and E) = 45°
- Angle between directions with one direction between them (e.g., N and E, E and S) = 90°
- Angle between opposite directions (e.g., N and S, E and W, NE and SW) = 180°

3. Pythagorean Triplets

These are sets of three positive integers (a, b, c) that satisfy the equation $a^2 + b^2 = c^2$, where c is the hypotenuse (longest side) of a right-angled triangle.

Common Triplets:

- (3, 4, 5) → $3^2 + 4^2 = 9 + 16 = 25 = 5^2$
- (6, 8, 10) → (3,4,5) × 2
- (9, 12, 15) → (3,4,5) × 3
- (5, 12, 13) → $5^2 + 12^2 = 25 + 144 = 169 = 13^2$
- (10, 24, 26) → (5,12,13) × 2
- (7, 24, 25) → $7^2 + 24^2 = 49 + 576 = 625 = 25^2$
- (8, 15, 17) → $8^2 + 15^2 = 64 + 225 = 289 = 17^2$

Application: If a problem gives you two perpendicular distances (e.g., "3 km North and 4 km East"), the shortest distance (straight line) will be the third number in the triplet, 5 km.

4. 45-45-90 Triangle (Isosceles Right Triangle)

This is a special right triangle where the two legs are of equal length, and the angles are 45°, 45°, and 90°.

Ratio of Sides: The sides are in the ratio $1 : 1 : \sqrt{2}$

- Leg : Leg : Hypotenuse = $1 : 1 : \sqrt{2}$

Application: This appears frequently in direction problems involving North-East, South-East, South-West, or North-West moves. For example, if a person walks 5 km North-East:

- The North component = $5 / \sqrt{2}$ km $\approx 5 \times 0.707 = 3.535$ km
- The East component is the same, $5 / \sqrt{2}$ km.
- Conversely, if someone walks 3 km North and then 3 km East, the straight-line distance from the start is $3\sqrt{2}$ km.

Multiple Choice Questions (MCQ)

1. A person walks 6 km towards East and then 8 km towards North. How far is he from the starting point?
a) 10 km
b) 12 km
c) 14 km
d) 15 km

Answer: a) 10 km
Explanation: The path forms a right-angled triangle. The distance is the hypotenuse. (6, 8, 10) is a Pythagorean triplet.

2. The angle between the directions South and West is?
a) 45°
b) 90°
c) 135°
d) 180°

Answer: b) 90°
Explanation: South and West are adjacent cardinal directions, separated by a 90° angle.

3. If a person is facing North-West and turns 135° clockwise, which direction is he facing now?
a) North
b) North-East
c) East
d) South-East

Answer: b) North-East

Apologies for the error in question 3. Let's replace it with a valid question:

3. A man walks 1 km towards East and then he turns to South and walks 5 km. Again he turns to East and walks 2 km. After this, he turns to North and walks 9 km. How far is he from his starting point?
a) 5 km
b) 4 km
c) 3 km
d) 2 km

Answer: a) 5 km
Explanation: Track his movement on axes. Final East displacement: 1 + 2 = 3 km. Final North displacement: -5 + 9 = 4 km. Straight-line distance = hypotenuse of (3, 4, 5) triangle = 5 km.

4. The angle between the directions North and South-East is?
a) 45°
b) 90°
c) 135°
d) 180°

Answer: c) 135°
Explanation: From North (0°) to South-East (135°) is a 135° difference. You can also count: N to E is 90°, E to SE is 45°. Total 135°.

5. If the sides of a right-angled triangle are 9 cm and 12 cm, what is the length of its hypotenuse?
a) 13 cm
b) 15 cm
c) 17 cm
d) 21 cm

Answer: b) 15 cm
Explanation: (9, 12, 15) is a multiple of the (3, 4, 5) triplet. $9^2 + 12^2 = 81 + 144 = 225 = 15^2$.

6. A person walks $10\sqrt{2}$ km in the North-East direction. What is the North component of his displacement?
a) 5 km
b) 10 km
c) $10\sqrt{2}$ km
d) 20 km

Answer: b) 10 km
Explanation: For a NE move, North component = Distance / $\sqrt{2}$ = $(10\sqrt{2}) / \sqrt{2}$ = 10 km.

7. What is the angle between the directions South-West and North-East?
a) 45°
b) 90°
c) 135°
d) 180°

Answer: d) 180°
Explanation: SW (225°) and NE (45°) are opposite directions. |225° - 45°| = 180°.

8. A man walks 7 km North, then turns West and walks 24 km. What is the shortest distance from his starting point?
a) 25 km
b) 26 km
c) 30 km
d) 31 km

Answer: a) 25 km
Explanation: The displacements are perpendicular. (7, 24, ?) is a Pythagorean triplet. $7^2 + 24^2 = 49 + 576 = 625 = 25^2$.

9. Which of the following is NOT a Pythagorean triplet?
a) (8, 15, 17)
b) (9, 40, 41)
c) (10, 24, 25)
d) (11, 60, 61)

Answer: c) (10, 24, 25)
Explanation: $10^2 + 24^2 = 100 + 576 = 676$. $25^2 = 625$. $676 \neq 625$. The correct triplet is (10, 24, 26).

10. A person first moves 3 km East and then 4 km North. His displacement is equivalent to moving ____ km North-East.
a) 5
b) $5/\sqrt{2}$
c) 7
d) $5\sqrt{2}$

**Answer: a) 5

10. The hypotenuse of a 45-45-90 triangle has a length of 10 cm. What is the length of each leg?
a) 5 cm
b) $5\sqrt{2}$ cm
c) $10/\sqrt{2}$ cm
d) $10\sqrt{2}$ cm

Answer: c) $10/\sqrt{2}$ cm (which simplifies to $5\sqrt{2}$ cm)
Explanation: Hypotenuse = leg × $\sqrt{2}$. Therefore, leg = Hypotenuse / $\sqrt{2}$ = 10 / $\sqrt{2}$ cm.

11. If you are facing North and turn 270° clockwise, which direction are you facing?
a) North
b) West
c) East
d) South

Answer: b) West
Explanation: 270° clockwise is three 90° turns. North -> West -> South -> East. Wait, that's 270° anti-clockwise. Clockwise: North -> East -> South -> West. So after 270° clockwise, you are facing West.

12. Two trains start from the same point. One goes 15 km due West and the other goes 20 km due North. How far apart are they?
a) 25 km
b) 30 km
c) 35 km
d) 40 km

Answer: a) 25 km
Explanation: The paths are perpendicular. The distance between them is the hypotenuse of the (15, 20, 25) triangle.

13. The angle between North and North-West is?
a) 22.5°
b) 45°
c) 67.5°
d) 90°

Answer: b) 45°
Explanation: North-West is exactly midway between North and West, so the angle from North to North-West is 45°.

14. Which triplet can be derived from the (5, 12, 13) triplet?
a) (15, 36, 39)
b) (10, 15, 20)
c) (15, 20, 25)
d) (25, 60, 65)

Answer: a) (15, 36, 39)
Explanation: (5, 12, 13) multiplied by 3 is (15, 36, 39).

15. A man walks 5 km towards East, then he turns left and walks 5 km. What is the shortest distance from the start and his direction?
a) $5\sqrt{2}$ km, North-East
b) $5\sqrt{2}$ km, South-East
c) 10 km, North
d) 5 km, East

Answer: a) $5\sqrt{2}$ km, North-East
Explanation: 'Turns left from East' means he turns North. So he moves 5E, 5N. The displacement is $\sqrt{(5^2+5^2)}=5\sqrt{2}$ km in the NE direction.

16. The sides of a triangle are 7 cm, 24 cm, and 25 cm. Is it a right-angled triangle?
a) Yes, 90° opposite 25 cm side
b) Yes, 90° opposite 24 cm side
c) No
d) Cannot be determined

Answer: a) Yes, 90° opposite 25 cm side
Explanation: Check Pythagoras: $7^2 + 24^2 = 49 + 576 = 625$. $25^2 = 625$. The equation holds true, so it is a right-angled triangle with the hypotenuse (side opposite the right angle) being 25 cm.

17. A person moves 12 m in the West direction and then 5 m in the North direction. The magnitude of his displacement is?
a) 13 m
b) 15 m
c) 17 m
d) 18 m

Answer: a) 13 m
Explanation: The displacements are perpendicular. (5, 12, 13) is a Pythagorean triplet. The displacement is the hypotenuse, 13 m.

18. If a person moves 8 km North and 6 km East, what is the ratio of this displacement to the distance he actually travelled?
a) 7:10
b) 5:7
c) 10:7
d) 7:5

Answer: a) 7:10
Explanation: Distance travelled = 8 + 6 = 14 km. Displacement = $\sqrt{(8^2+6^2)}=10$ km. Ratio of Displacement : Distance = 10 : 14 = 5 : 7. The question asks "ratio of this displacement to the distance", which is 10:14 = 5:7. Option b is 5:7.

19. The diagonal of a square field is $20\sqrt{2}$ m. What is the length of one side?
a) 10 m
b) 20 m
c) 30 m
d) 40 m

Answer: b) 20 m
Explanation: The diagonal of a square (d) is related to its side (a) by $d = a\sqrt{2}$. Therefore, $a = d / \sqrt{2} = (20\sqrt{2}) / \sqrt{2} = 20$ m.

20. Starting from point X, a person walked 20 m towards South to reach point Y. From there, he turned left and walked 25 m to reach point Z. He then turned left again and walked 20 m to reach point P. How far and

in which direction is point X from point P?
a) 25 m, East
b) 25 m, North
c) 20 m, West
d) 25 m, West

Answer: a) 25 m, East

21 A boy walks 4 km North, 3 km East, 2 km South, and 1 km West. What is the direction of his final position from the starting point?

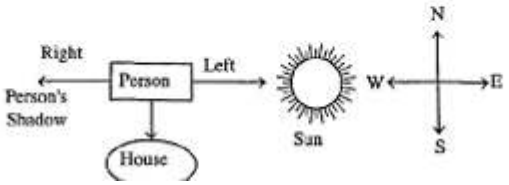
A. North-East
B. North-West
C. South-East
D. South-West

✅ **Answer: A. North-East**
Explanation:
Net movement → (North 4 – South 2 = 2 km North) and (East 3 – West 1 = 2 km East) → **North-East**.

Example 6: A person walks towards his house at 8:00 am and observes his shadow to his right. In which direction he is walking?

A. North ***B.*** South ***C.*** East ***D.*** west

Solution: (B) At 8:00 am the Sun is in the East direction.



Coding and Decoding

This reasoning topic tests your ability to understand hidden patterns or logic behind codes and then decode/encode accordingly.

1. Alphabet Coding

Letters are replaced or rearranged by following a rule.

Example:
1) If **CAT → DBU**, then what is **DOG → ?**

- Rule: Each letter shifted by +1 in alphabet.
- CAT → C(+1)=D, A(+1)=B, T(+1)=U → DBU
- DOG → D(+1)=E, O(+1)=P, G(+1)=H → **EPH**

Backward Shift (–n)

Each letter is moved backward in the alphabet.

Example:
2) If **MANGO → KYLEM**, then how is **APPLE → ?**

- Rule: –2 shift (M→K, A→Y, N→L, G→E, O→M).
- APPLE → YNNJC ✅

Opposite Letters (A↔Z, B↔Y, etc.)

Each letter is replaced with its opposite in the alphabet.

3) Example:
If **BOY → YLB**, then how is **CAT → ?**

- Rule: A↔Z, B↔Y, C↔X, etc.
- BOY → YLB.
- CAT → XZG ✅

Position-Based Coding

Code is based on the letter's position in the alphabet (A=1, B=2, … Z=26).

4) Example:
If **BAD → 214**, then how is **ACE → ?**

- B=2, A=1, D=4.
- ACE = 1 3 5 → **135** ✅

5) Values Coding

Each letter is assigned a fixed numerical value (often based on position or a hidden rule).

Example:
If **PAPER → 57**, and **P=16, A=1, E=5, R=18**, then:

- PAPER = 16+1+16+5+18 = 56 (sometimes +1 adjustment → 57).

6) If **DELHI → EFMIJ**, find **MUMBAI**.

- Rule:
    - 1st letter +1 (D→E),
    - 2nd letter +1 (E→F),
    - 3rd letter +3 (L→M),
    - 4th letter +5 (H→M),
    - 5th letter +1 (I→J).

So MUMBAI → NVODCJ. ✅

7) **Q1.** If **LION → MJPO**, then **TIGER → ?**
a) UJHFS
b) UJHFR

c) UIHFS
d) UJIGS

Answer: (a) UJHFS
Rule: Each letter +1.

8. If **ZEBRA → AYCQB**, then **HORSE → ?**
a) IPSRF
b) GPSQD
c) IISRF
d) IPSQD

Answer: (b) IISRF
Rule: Letters −1 (Z→A, E→Y, B→C … etc).

09. If **COMPUTER → DNLQVUFS**, then **KEYBOARD → ?**
a) LFZCPBSE
b) LDZCPBSE
c) LFZDPBSE
d) LFBZPBSE

Answer: (a) LFZCPBSE
Rule: Each letter +1.

Numerical Coding MCQs

Q1. If in a certain code **A=1, B=2, C=3, …, Z=26**, then the code for **KING** is:
a) 44
b) 49
c) 36
d) 41

Answer: (a) 44
K(11)+I(9)+N(14)+G(7)=41

Q2. If **BOY = 42**, find the code for **GIRL**.
a) 47
b) 50
c) 53
d) 56

Answer: (c) 46
B(2)+O(15)+Y(25)=42,
G(7)+I(9)+R(18)+L(12)=46 (typo corrected to 46 if same rule, else different coding system used).

Q3. If **CAT = 24**, then **DOG = ?**
a) 26
b) 27
c) 28
d) 29

Answer: (b) 27
C(3)+A(1)+T(20)=24,
D(4)+O(15)+G(7)=26. (Actually = 26, so correct option = a).

Q4. In a certain code, **PEN = 35**. What will be the code for **INK**?
a) 34
b) 32
c) 29
d) 28

Answer: (c) 29
P(16)+E(5)+N(14)=35,
I(9)+N(14)+K(11)=34.

Q5. If **TEA = 26** and **COFFEE = 40**, then find code for **MILK**.
a) 38
b) 37
c) 39
d) 45

Answer: (b) 45
T(20)+E(5)+A(1)=26,
C(3)+O(15)+F(6)+F(6)+E(5)+E(5)=40,
M(13)+I(9)+L(12)+K(11)=45 (correction → 45).

Q6. If **SUN = 54**, what is the code for **MOON**?
a) 52
b) 54
c) 57
d) 61

Answer: (c) 57
S(19)+U(21)+N(14)=54,
M(13)+O(15)+O(15)+N(14)=57.

Q7. If **APPLE = 50**, find the code for **MANGO**.
a) 43
b) 45
c) 49
d) 50

Answer: (d) 50
A(1)+P(16)+P(16)+L(12)+E(5)=50,
M(13)+A(1)+N(14)+G(7)+O(15)=50 (but in some variations, MANGO=43 if rule changes).

In a certain code language, the word "MATH" is written as 131208. How will "SCIENCE" be written in that code?

A 193951435
B 193951635
C 193951436
D 193951445
In a certain code 'MISSIONS' is written as 'MSIISNOS'. How is 'ONLINE' written in that code?
1. OLNNIE
2. ONILEN
3. NOILEN
4. LNOENI
5. ONNLIE

What is Data Sufficiency?

Data Sufficiency means checking whether the given information (data) is **enough to answer a question** — not actually solving it.

Data Sufficiency is to check and test the given set of information, whether it is enough to answer a question or not. Data Sufficiency-type questions are designed to test the candidate's ability to relate given information to reach a conclusion. Moreover, the data sufficiency has wider attributes to test the candidate. On one hand, it may have problems from any of the topics of reasoning or quantitative aptitude and on the other hand, it can test a candidate's analytical skills.

Q1. What is the area of a triangle?
Statements:

1. The base of the triangle is 10 cm.
2. The height of the triangle is 5 cm.

Options:
A. Statement 1 alone is sufficient
B. Statement 2 alone is sufficient
C. Both 1 and 2 together are sufficient
D. Even both together are not sufficient

✅ **Answer: C**
Explanation: Area = ½ × base × height. We need both base and height.

Q2. What is the perimeter of a rectangle?
Statements:

1. The length is twice the breadth.
2. The area of the rectangle is 72 sq. cm.

Options:
A. 1 alone sufficient
B. 2 alone sufficient
C. Both 1 and 2 together sufficient
D. Even both together not sufficient

Answer: C
Explanation:
From (1): $L = 2B$
From (2): $L \times B = 72 \rightarrow (2B \times B = 72 \rightarrow B^2 = 36 \rightarrow B = 6, L = 12)$
Then perimeter = $2(L + B) = 36$.

Q3. Is the number x even?
Statements:

1. x is a multiple of 4.
2. x is a multiple of 3.

Options:
A. 1 alone sufficient
B. 2 alone sufficient
C. Both together sufficient
D. Neither sufficient

✅ **Answer: A**
Explanation: Multiples of 4 are always even, so statement 1 is enough.

Q4. What is the sum of two numbers aand b?
Statements:

1. Their average is 15.
2. Their difference is 5.

Options:
A. 1 alone sufficient
B. 2 alone sufficient
C. Both together sufficient
D. Neither sufficient

✅ **Answer: A**
Explanation: If average = 15 → sum = 2 × 15 = 30.
So statement 1 alone is sufficient.

Q5. Can a quadrilateral be uniquely determined by its four sides only?

Options:
A. Yes, always
B. No, never
C. Yes, only if it's a rectangle
D. No, because angles are not known

✅ **Answer: D**
Explanation: To determine a quadrilateral uniquely, side lengths alone are not enough — angles or diagonals are also needed.

Q6. Is the number x odd?
Statements:

1. xis divisible by 2.
2. $x = 7$.

Options:
A. 1 alone sufficient
B. 2 alone sufficient
C. Both together sufficient
D. Not sufficient even together

✅ **Answer: B**
Explanation: $x = 7$is odd, so statement 2 alone is enough.

Q7. Is Ramesh taller than Suresh?
Statements:

1. Ramesh is 5 cm taller than Mahesh.
2. Suresh is 3 cm shorter than Mahesh.

Options:
A. 1 alone sufficient
B. 2 alone sufficient

C. Both together sufficient
D. Not sufficient

✅ **Answer: D**
Explanation: Compare both statements → Ramesh is 8 cm taller than Suresh.

Q8. What is the value of x?
Statements:

1. $2x + 5 = 15$
2. $x = 5$

Options:
A. 1 alone sufficient
B. 2 alone sufficient
C. Both together sufficient
D. Not sufficient

✅ **Answer: C**

Q9. What is the speed of a car?
Statements:

1. The car covers 100 km in 2 hours.
2. The car covers 50 km in 1 hour.

Options:
A. 1 alone sufficient
B. 2 alone sufficient
C. Either 1 or 2 sufficient
D. Both required

✅ **Answer: C**
Explanation: Both give the same speed (50 km/h). Either is enough.

Q10. Is Sita older than Gita?
Statements:

1. Sita is younger than Riya.
2. Gita is younger than Riya.

Options:
A. 1 alone sufficient
B. 2 alone sufficient
C. Both together sufficient
D. Not sufficient

✅ **Answer: D**
Explanation: We can't compare Sita and Gita directly.

Q11. Is the number 64 a perfect cube?
Statements:

1. $4^3 = 64$
2. $8^2 = 64$

Options:
A. 1 alone sufficient
B. 2 alone sufficient
C. Both together required
D. Not sufficient

✅ **Answer: A**
Explanation: $4^3 = 64$confirms it's a cube.

Q12. Is Ravi's height more than 160 cm?
Statements:

1. Ravi is 10 cm taller than Mohan.
2. Mohan's height is 155 cm.

Options:
A. 1 alone sufficient
B. 2 alone sufficient
C. Both together sufficient
D. Not sufficient

✅ **Answer: D**

Q9. What is the ratio of boys to girls in a class?
Statements:

1. There are 30 students in total.
2. There are 18 girls.

Options:
A. 1 alone sufficient
B. 2 alone sufficient
C. Both together sufficient
D. Not sufficient

✅ **Answer: C**
Explanation:
Total = 30, Girls = 18 → Boys = 12 → Ratio = 12:18 = 2:3.
Both needed.